Jyothir Anurag Vasu Chinnaboina

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Education

University of Wisconsin–Madison – *B.S. in Computer Science and Data Science* — Madison, WI
Awards: Dean's List (4 semesters) — Expected May 2026
Coursework: Operating Systems, Algorithms, AI/ML, Statistics, Database Management, Data Structures — **GPA: 3.7/4.0**

Experience

Software Developer Intern (Part-time) – *Vijay Rekha Life Sciences* — **Aug 2025 – Present**
Sole Developer · vijayrekha.chinnaboina.com

- Automated 3+ hours of manual work into a single-click data extraction pipeline (Python, pandas, xlsxwriter) that transforms raw mass spectrometry data for 40+ biomedical compounds into formatted Excel reports.
- Engineered a PostgreSQL-backed data system (FastAPI, pandas) handling 300+ samples/day with validation and abnormality flagging across 30+ compounds.
- Accelerated clinical report validation by 85% using a batch processing pipeline (Python, pdfplumber, regex) that extracts and cross-validates patient data from 100+ PDF reports across multiple formats.

Mainframe Intern (Part-time) – *Wisconsin Department of Administration* — **Jun 2025 – Present**

- Reduced manual triage time by 90% with a Python data pipeline that parsed system logs into structured CSV datasets, flagged remediation status, and routed follow-ups across 5+ teams.
- Eliminated 4+ hours/week of manual reporting via a Python report orchestrator with YAML-driven configuration, JSON Schema validation, Jinja2 templating, and automated email distribution.
- Automated extraction of HTML configuration reports into structured Excel workbooks (Python, BeautifulSoup, xlsxwriter), processing 44 service classes across 6 policies with forward-fill data normalization.

ERP Integration Intern – *United States Pharmacopeia* — **Jun – Aug 2024**

- Improved data retrieval time by 70% with an Oracle ERP analytics tool (Python, oracledb, pandas) using optimized SQL queries joining 5 tables to surface order hold trends.
- Automated quarterly stakeholder reports, previously 1 week of manual work, with an ETL pipeline (pandas, Matplotlib, Seaborn) rendering visualizations into Word, PPTX, and PDF for executive distribution.
- Built a web scraping tool (Python, Selenium) that extracted and structured compliance data from internal portals, replacing manual copy-paste workflows across 3 cross-departmental teams.

Projects

Office Hours Queue | *Python, FastAPI, PostgreSQL, Redis, Docker* — **Sep 2025 – Present**
Co-Founder · *Advisor: Prof. Louis Oliphant (CS)* · ohq.cs.wisc.edu

- Reduced wait 1–3hr to 15min for 500+ students via a data-driven queue platform with PostgreSQL-backed state and ZSET-tiered ordering.
- Built a data ingestion pipeline indexing content from 4 sources into a unified ChromaDB vector store with metadata filtering, batch embedding, and per-document fallback on failure.
- Decreased average answer lookup time to ~2s with semantic search (FastAPI, ChromaDB, scikit-learn) across 4 knowledge sources and DBSCAN/KMeans clustering for pattern detection.
- Enabled one-command deployments for 6 services by containerizing with Docker Compose and automating CI/CD via GitHub Actions and Portainer webhooks.

E-Beam 3D Printer Control System | *Python, Node.js, Express* — **Jan – Aug 2025**
Morgridge Institute for Research (Prof. Kevin Eliceiri)

- Architected a multi-threaded Python data ingestion system processing telemetry from 6+ subsystems with fault-tolerant backoff and graceful degradation.
- Built a real-time data streaming pipeline (Node.js, Express) with log2-based downsampling to aggregate weeks of sensor data across 6 temperature sensors and 13 interlock points for remote analysis.

Supply Chain Forecasting Model | *Python, pandas, scikit-learn* — **Jun – Aug 2024**
United States Pharmacopeia

- Built a regression-based forecasting model (scikit-learn, pandas) predicting medicine supply chain shortages from 2 years of historical order data across 50+ product lines.
- Engineered the feature pipeline (lag windows, rolling averages, holiday encodings) and tuned regressor hyperparameters via cross-validation.

Skills

Languages: Python, SQL, Java, Bash, HTML/CSS
Data & ML: pandas, NumPy, scikit-learn, ChromaDB, Matplotlib, Seaborn
Pipelines & ETL: ETL Pipelines, Data Validation, Web Scraping (BeautifulSoup, Selenium), pdfplumber, JSON Schema, Jinja2
Frameworks & Tools: Flask, FastAPI, Pydantic, Uvicorn, Docker, Docker Compose, Git, GitHub Actions, Pytest
Databases & Cloud: PostgreSQL, Oracle DB, MySQL, SQLite, Redis, Google Cloud Platform, Linux